

Experiment No: 3

Title: Implement control structures and form validation; create a grading system based on user-entered marks.

Software/Tools Required:

Visual Studio Code (or any text editor), Web Browser (Google Chrome, Microsoft Edge, Firefox, etc.)

Theory:

JavaScript is a powerful client-side scripting language used to make web pages interactive and dynamic. It provides various control structures, loops, and form validation techniques that allow developers to process user input, make logical decisions, and display dynamic results without reloading the webpage. These features are essential for developing responsive and user-friendly web applications.

1. if-else Statement

- Used to make decisions based on conditions.
- Executes different blocks of code depending on whether the condition is true or false.

2. switch Statement

- Used when multiple conditions need to be checked.
- Executes the matching case based on the given expression.

3. Loops

- **for:** Repeats code for a fixed number of iterations.
- **while:** Repeats code while a condition is true.
- **do-while:** Executes the code at least once before checking the condition.

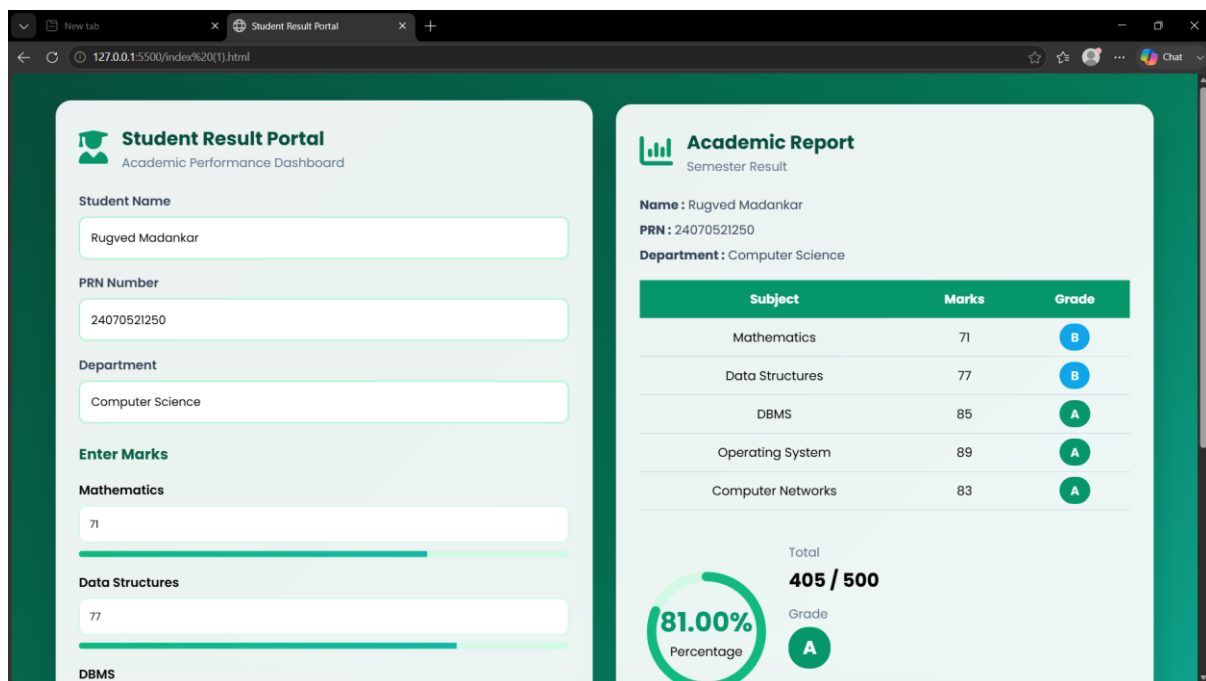
4. Form Validation

- Ensures that user inputs are valid before processing.
- Checks for empty fields, valid ranges, and correct data types.

Aim of the CASE STUDY: The Student Grading System is a JavaScript-based web application that demonstrates the use of control structures and form validation. The application accepts student information, validates the entered data, calculates academic performance, and displays the final result dynamically.

Working of the System:

1. The user enters the student's name, roll number, and marks.
2. The program validates every input field.
3. If any field is empty or contains invalid data, an error message is displayed.
4. The final result is displayed on the webpage without refreshing it.



Result/Conclusion:

The experiment was successfully completed. The concepts of var, let, and const were understood and implemented. Template literals were used to display formatted output, and destructuring simplified data extraction from objects. A billing calculator was successfully developed to accept user input, calculate the bill with GST, and display the final amount accurately.

Output:

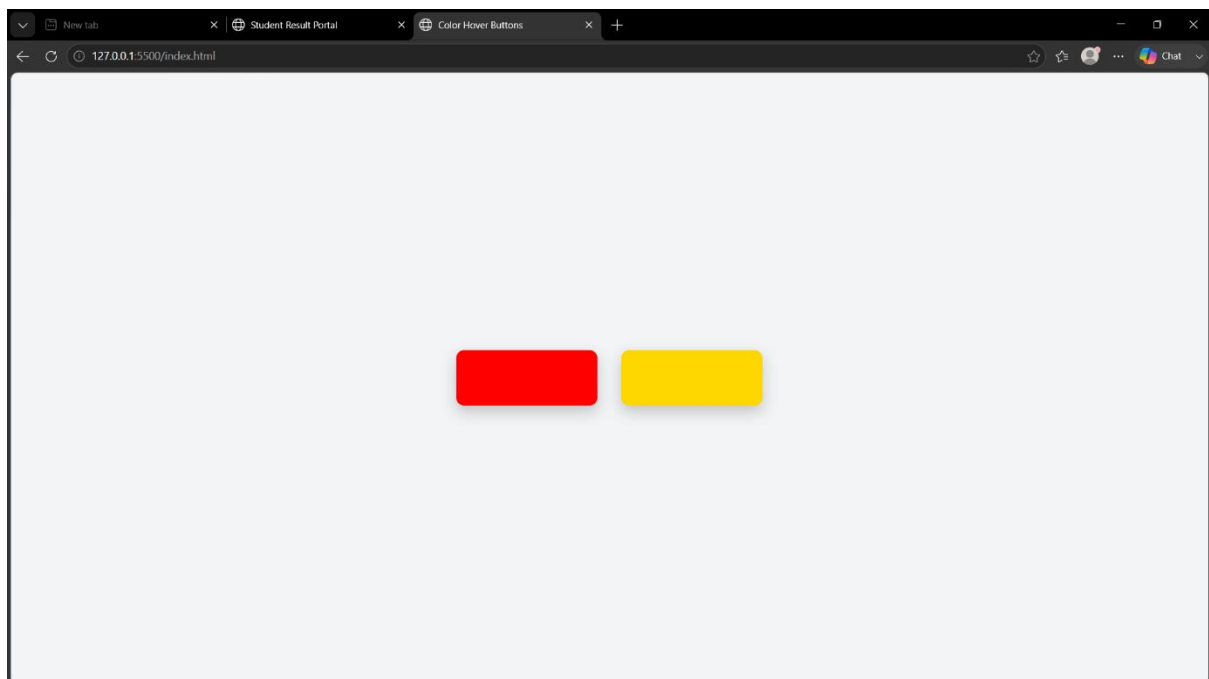
After running the program in a web browser, a form is displayed where the user enters the student's details and marks obtained in different subjects. Once the Submit or Calculate button is clicked, the JavaScript program first validates all the entered information. If any field is left blank or marks are entered outside the valid range (0–100), the application displays an appropriate validation message and prevents further processing until the errors are corrected.

If all the entered values are valid, the application performs the following operations:

- Calculates the total marks obtained by the student.
- Computes the overall percentage.
- Determines the grade according to the predefined grading criteria.
- Displays the student's details, total marks, percentage, grade, and pass/fail status dynamically on the webpage.

The output is generated instantly without reloading the page, demonstrating the dynamic nature of JavaScript.

Additional Assignment:



CONCLUSION: This experiment demonstrated JavaScript control structures, form validation, and event handling. A Student Grading System was developed to calculate percentage, assign grades, and display results dynamically. A box color swap on mouse hover was also implemented.